Collection Partnership

Smoker Status Prediction Challenge

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Project Overview

Identifying health risk factors using biological signals is a core application of data analytics and artificial intelligence in medicine.

The goal of this project is to build a machine learning model that can accurately predict whether a person is a smoker (1) or a non-smoker (0).

(1 for Smoker, 0 for Non-Smoker) based on physical attributes, cardiovascular metrics, and laboratory test results.

Key insights

The Goal: The project uses simple health data (like height, weight, and blood tests)

to guess whether a person is a smoker or a non-smoker.

Machine learning models used

- Logistic Regression
- Random Forest
- LightGBM
- XGBoost

XG boost (Extreme Gradient Boosting)

Highly accurate, machine learning model

Made out of multiple small decision trees one after the other

Made to specifically correct/fix the errors made by previous tree

XG boost (Extreme Gradient Boosting)

Advantages

Extremely accurate

Predicting

Build-in smart protections

Outliers
Missing data
Prevents overfitting

Disadvantages

High Memory consumption

Hyperparameter complexity

Resource Intensive on Scaling

XG boost (Extreme Gradient Boosting)

Used in the code

Model Selection & Evaluation phase

The Supervised Learning Pipeline

Stratified 5-Fold Cross-Validation

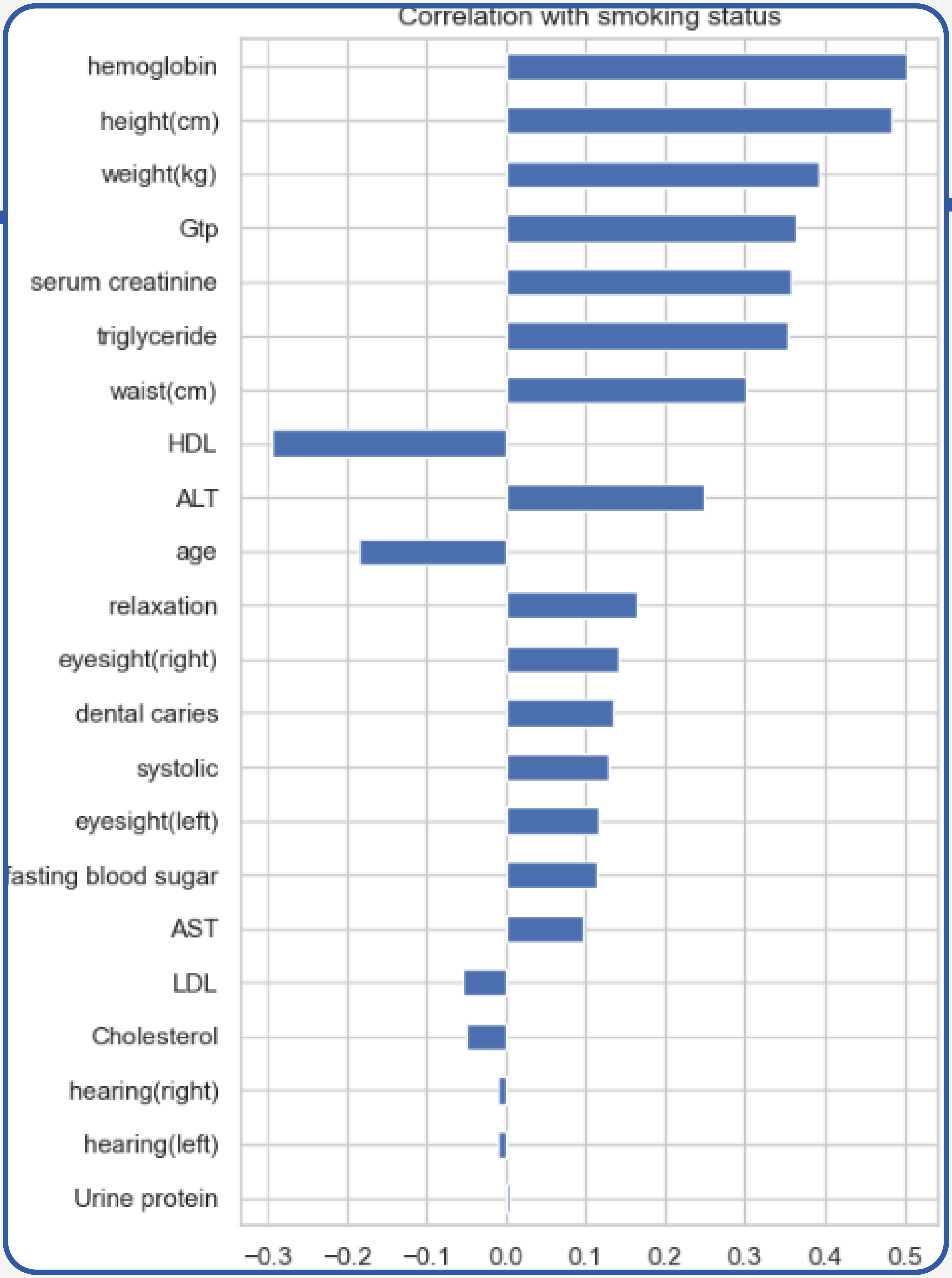
Probability Optimization (predict_proba)

Models

Exploratory Data Analysis (EDA)

Top 3 Predictors

Hemoglobin
Height (cm)
Weight (kg)

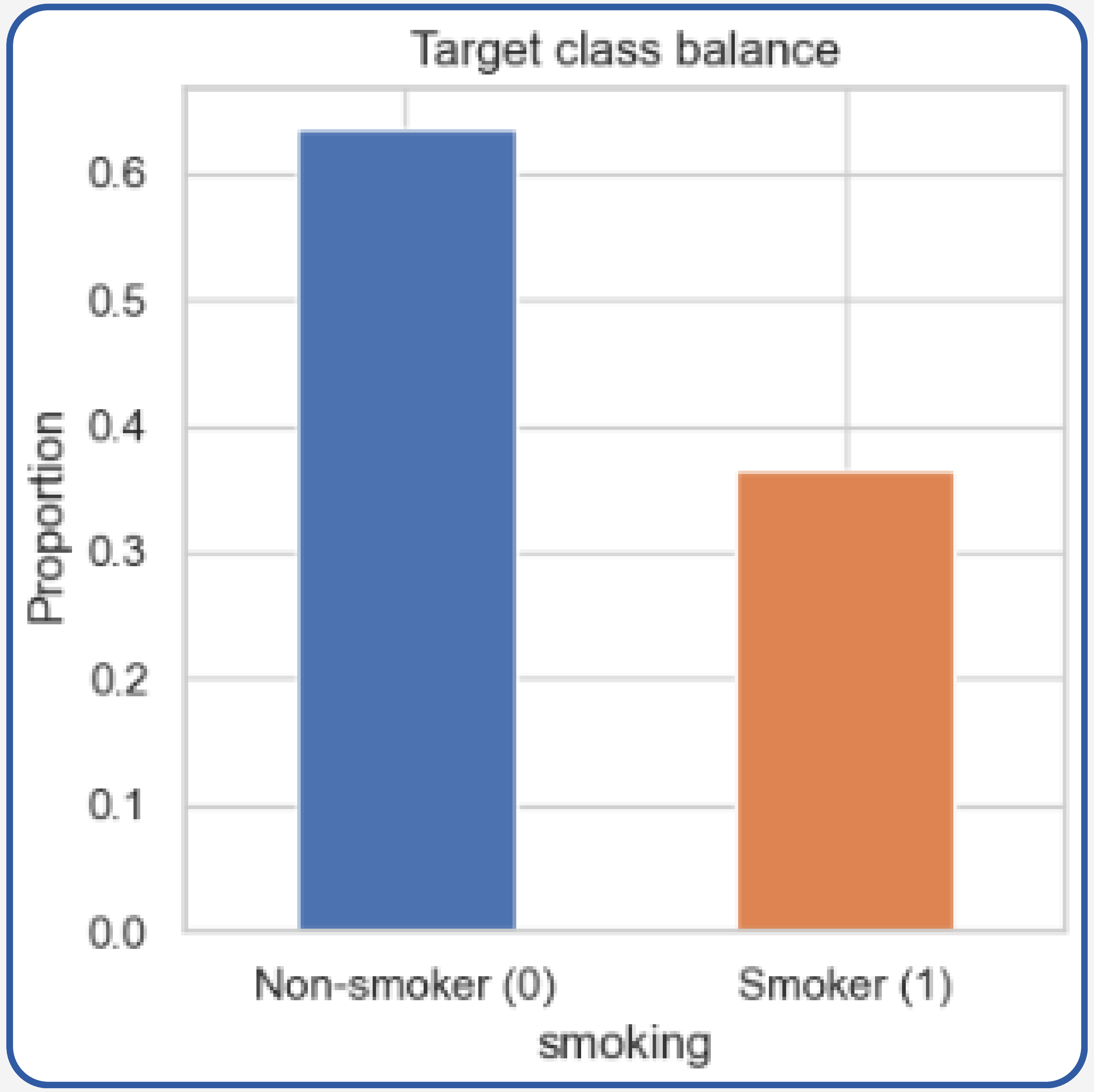


Models

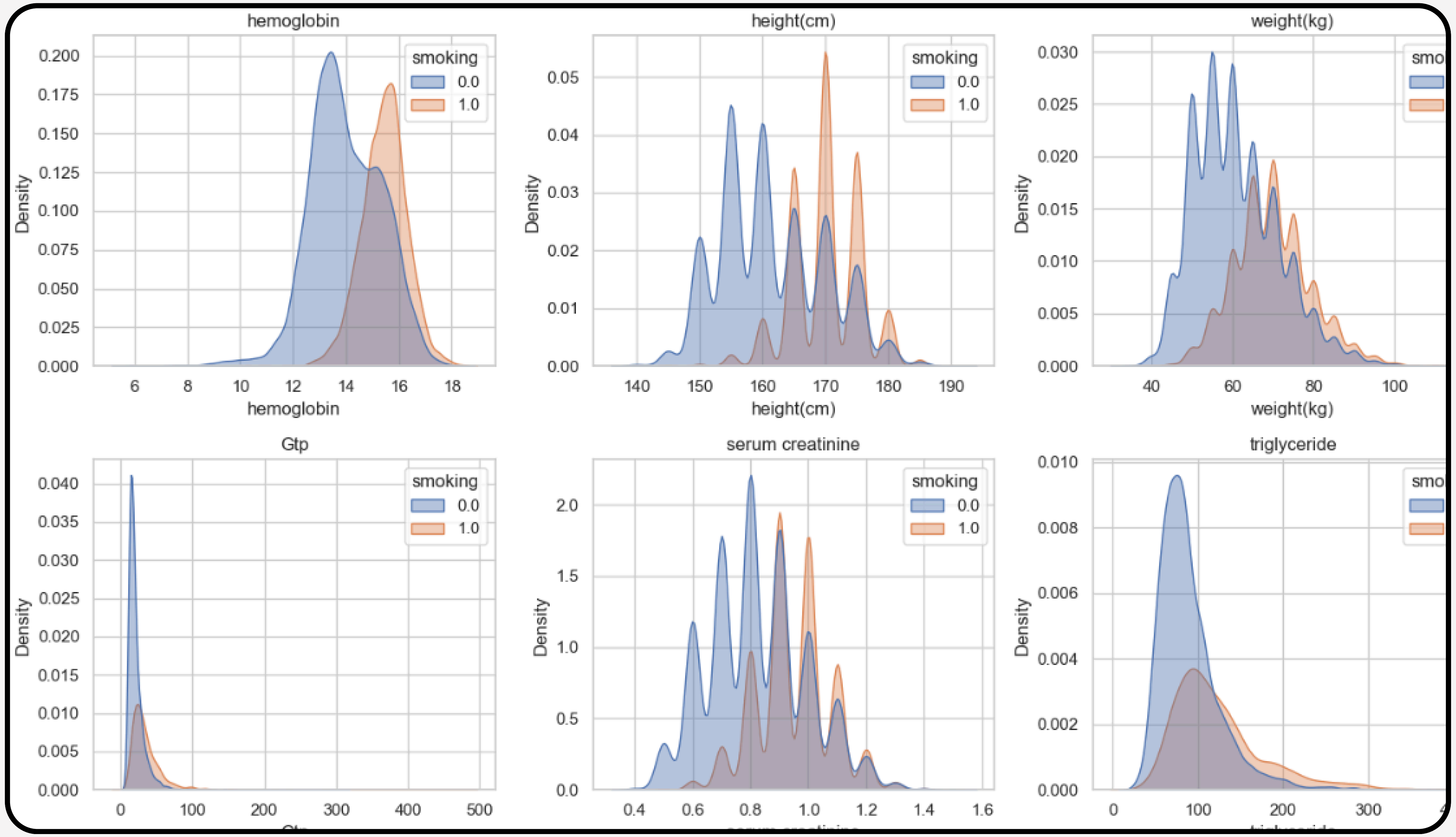
Non-smokers (0): 63%

Smokers (1): 37%

Dataset is mildly imbalanced



Models (KDE)



hemoglobin
height(cm)
weight(kg)
Gtp
serum creatinine
triglyceride
smoking
0.0
1.0
Density

Thank you